Software Requirements Specification

for

Digital Wallet Management System

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Digital Wallet Management System (DWMS), Version 1.0. It describes the complete set of requirements for the initial release of the product and serves as the primary reference for design, implementation, testing, and acceptance. This document covers the full scope of the DWMS — wallet management, money transfer, payment processing, transaction history, notifications, and administration — and does not specify requirements for the internal implementation of external banking or payment-gateway systems, only the interfaces the DWMS consumes from them.

1.2 Document Conventions

This document follows the IEEE Std 830-1998 recommended structure for software requirements specifications. Each functional requirement is uniquely identified with a tag of the form REQ-<feature>-<number> (for example, REQ-XFER-2) so that it can be traced through design, implementation, and test artifacts. The word “shall” denotes a mandatory capability; “should” denotes a recommended but non-mandatory capability. Unless a specific priority is stated for an individual requirement, it inherits the priority declared for its parent system feature in Section 4. Placeholders that still require stakeholder decisions are marked “TBD” and are collected in Appendix C.

1.3 Intended Audience and Reading Suggestions

This document is intended for four audiences: software developers and engineers implementing the system; project supervisors and reviewers assessing scope, design, and progress; quality assurance engineers deriving test cases and acceptance criteria; and technical writers producing end-user documentation. Readers unfamiliar with the project should begin with Section 2 (Overall Description) for a high-level orientation, then proceed to Section 3 (External Interface Requirements) and Section 4 (System Features) for detailed, testable requirements. Section 5 should be read closely by anyone responsible for performance, security, reliability, or other quality attributes. The appendices provide supporting reference material, including a glossary and analysis diagrams.

1.4 Product Scope

The Digital Wallet Management System is a software application that enables users to securely manage digital funds and perform financial activities, including registering and verifying an account, viewing wallet balances, adding money through supported funding methods, transferring funds to other users, initiating and confirming payments, reviewing transaction history, and receiving notifications about account activity. Administrators are provided with tools to monitor users and transactions and to respond to suspicious activity. The system's primary objective is to provide a secure, reliable, scalable, and easy-to-use platform for managing digital financial transactions while maintaining accurate wallet balances and complete, auditable transaction records. The DWMS integrates with external banks and payment gateways to validate and process supported financial operations but does not itself implement those external services.

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.

- Digital Wallet Management System — Software Engineering Design Document (internal project document), Version 1.0, prepared by Surya.
- Applicable payment-industry data-security guidance (e.g., PCI DSS) governing storage and handling of financial and payment data — reference to be finalized (see Appendix C).

2. Overall Description

2.1 Product Perspective

The DWMS is a new, self-contained product rather than a follow-on to an existing system. It is organized into a presentation layer (the web/mobile client), an application layer implementing the core business logic (authentication, wallet, transaction, payment, and notification services), a data layer that persists users, wallets, the transaction ledger, and audit logs, and an integration layer that connects to external bank APIs, payment gateways, and OTP/notification services. This layered separation is intended to improve maintainability, testability, and scalability. Figure 1 in Appendix B illustrates the system context, and Figure 2 illustrates the layered architecture.

2.2 Product Functions

At a high level, the DWMS shall allow users to:

- Register and verify a new account.
- Authenticate securely to access their wallet.
- View wallet balance and manage basic wallet settings.
- Add money to their wallet through a supported funding method.
- Transfer money to other users.
- Initiate and confirm payments to merchants or third parties.
- View and filter transaction history.
- Receive notifications about account and transaction events.

Administrators shall additionally be able to monitor users and transactions and to investigate and act on suspicious activity. Figure 3 in Appendix B (Use Case Diagram) shows how these functions map to the User, Administrator, and Bank/Payment Service actors.

2.3 User Classes and Characteristics

- User / Customer — the primary user class. Registers, authenticates, and performs wallet, transfer, payment, and history operations. Assumed to have ordinary digital literacy but no technical expertise; this is the most important class for the system to satisfy.
- Administrator — an internal, trusted user class with elevated privileges to monitor users and transactions and investigate suspicious activity. Assumed to have domain training in compliance and fraud-review procedures.
- Bank / Payment Gateway (external system actor) — not a human user, but an external system that validates and processes supported financial operations on the DWMS's behalf.

2.4 Operating Environment

The DWMS shall operate as a client–server web/mobile application. The client shall run on current versions of major web browsers and/or a mobile operating system (iOS/Android); the server-side components shall run on a scalable, cloud-hostable backend environment. The system shall coexist with,

and depend on the availability of, external bank and payment-gateway services and an OTP/notification provider.

TBD — specific supported OS/browser versions and the target cloud hosting environment (Section 2.4).

2.5 Design and Implementation Constraints

- All financial operations shall be designed to avoid partial updates to wallet balances; a failed transaction shall not leave the ledger in an inconsistent state.
- Security-by-design: authentication, authorization, and input validation shall be incorporated from the outset of design rather than added later.
- The system shall be built using a modular, layered architecture (presentation, application, data, integration) to support low coupling and high cohesion.
- Sensitive financial and personal data shall be handled according to applicable security and regulatory requirements (e.g., encryption at rest, PCI-relevant handling of payment data).
- Network connectivity is required for all online transactions; DWMS availability is therefore bounded by the availability of external banking/payment services.

2.6 User Documentation

The following user documentation shall be delivered with the product: a user manual/help guide covering registration, wallet management, transfers, payments, and transaction history; in-app contextual help and error-message guidance; and an administrator guide covering monitoring and suspicious-activity handling.

TBD — delivery format for user documentation (online help vs. downloadable PDF) (Section 2.6).

2.7 Assumptions and Dependencies

- The system assumes continuous or near-continuous network connectivity for online transactions.
- The system depends on third-party banks and payment gateways to validate and process certain financial operations; their availability and processing time may affect DWMS availability and performance.
- The system depends on an OTP/notification service for account verification and transaction alerts.
- Transaction limits are assumed to depend on a user's verification status and on configurable business rules that may change independently of this SRS.

3. External Interface Requirements

3.1 User Interfaces

The DWMS shall provide a web and/or mobile user interface comprising: a registration/login screen; a dashboard showing wallet balance and recent activity; screens for adding money, transferring money, and making payments, each requiring an explicit confirmation step before an irreversible financial action is committed; a transaction-history screen supporting filtering; and a notifications view. The interface shall present clear, understandable error messages for invalid input and failed operations, and shall provide consistent navigation and standard controls (e.g., help, back, confirm/cancel) across all screens. Detailed screen layouts are to be documented in a separate user-interface specification.

3.2 Hardware Interfaces

The DWMS does not require dedicated hardware. It shall operate on standard user devices — desktop/laptop browsers and smartphones/tablets — capable of running the supported client. No direct control of hardware peripherals is required beyond standard device I/O (for example, a device camera for identity-verification document capture, if that capability is included).

3.3 Software Interfaces

- Bank / Payment Gateway APIs — used to validate and process funding, transfers, and payments; data exchanged includes authorization requests/responses and transaction status.
- OTP / Notification Service — used to deliver one-time passwords for identity verification and to push account/transaction notifications.
- Database — the persistent data store for users, wallets, the transaction ledger, and audit logs (see Sections 2.1 and 7 of the accompanying design document).

TBD — specific vendor products and versions for the database, payment gateway, and notification service (Section 3.3).

3.4 Communications Interfaces

All communications between the client and server, and between the DWMS and external banks/payment gateways, shall use HTTPS (TLS) to protect data in transit. The system shall use a request/response protocol (e.g., REST over HTTPS) for client–server communication and for integration with external services. Notification delivery may use push notification, SMS, or email channels as configured.

TBD — specific message formats and required throughput/rate limits for external communications (Section 3.4).

4. System Features

This section organizes the functional requirements of the DWMS by system feature, the major services provided by the product. Each feature is described with its priority, its stimulus/response behaviour, and the individually traceable functional requirements that implement it.

4.1 User Registration and Authentication

4.1.1 Description and Priority

Allows a new user to create and verify an account, and allows a returning user to authenticate securely before accessing wallet functions. Priority: High.

4.1.2 Stimulus/Response Sequences

- User enters registration details → system validates the input and creates credentials → system sends a verification code (e.g., OTP/email) → user verifies → account is created and made accessible.
- User enters login credentials → system validates them → user is granted access to the dashboard, or shown an error message if the credentials are invalid.

4.1.3 Functional Requirements

REQ-REG-1: The system shall allow a user with no existing account to register by providing the required identity and contact details.

REQ-REG-2: The system shall validate submitted registration details and reject incomplete or invalid submissions with a clear, specific error message.

REQ-REG-3: The system shall require account verification (e.g., OTP or email confirmation) before granting full access to wallet functions.

REQ-AUTH-1: The system shall authenticate a user's credentials (e.g., username/password) before granting access to any wallet function.

REQ-AUTH-2: The system shall protect an account against repeated failed authentication attempts (TBD: specific lockout threshold, see Appendix C).

REQ-AUTH-3: The system shall support secure session management, including automatic session expiration after a period of inactivity (TBD: timeout value, see Appendix C).

4.2 Wallet Management

4.2.1 Description and Priority

Allows an authenticated user to view their wallet balance and add money to it through a supported funding method. Priority: High.

4.2.2 Stimulus/Response Sequences

- User selects “Add Money” → enters an amount → chooses a funding method → confirms → system processes the request with the bank/payment gateway → system updates the wallet balance and ledger.

4.2.3 Functional Requirements

REQ-WAL-1: The system shall create a wallet automatically upon successful account verification.

REQ-WAL-2: The system shall display the current wallet balance to an authenticated user.

REQ-WAL-3: The system shall allow a user to add money to their wallet via a supported funding method.

REQ-WAL-4: The system shall validate a funding request and require explicit user confirmation before processing it.

REQ-WAL-5: The system shall update the wallet balance only after the funding transaction is confirmed successful by the bank/payment gateway.

4.3 Money Transfer

4.3.1 Description and Priority

Allows an authenticated user with sufficient balance to send money to another user of the system. Priority: High.

4.3.2 Stimulus/Response Sequences

- User authenticates → enters recipient and amount → system validates the recipient and the sender's balance → user confirms → system processes the transaction → system updates the wallet/ledger for both parties → system issues a receipt, notification, and audit-log entry.

4.3.3 Functional Requirements

REQ-XFER-1: The system shall allow an authenticated user to specify a recipient and an amount to transfer.

REQ-XFER-2: The system shall validate that the recipient exists and that the sender's balance is sufficient before proceeding.

REQ-XFER-3: The system shall require explicit user confirmation before committing a transfer.

REQ-XFER-4: The system shall update the sender's and recipient's wallet balances atomically; a failed transfer shall not leave either balance in a partially updated state.

REQ-XFER-5: The system shall record every transfer in the transaction ledger and audit log, and shall notify both the sender and the recipient.

4.4 Payment Processing

4.4.1 Description and Priority

Allows an authenticated user to initiate and confirm a payment to a merchant or other supported payee. Priority: High.

4.4.2 Stimulus/Response Sequences

- User initiates a payment → enters amount and payee details → system validates the request → user confirms → system processes the payment with the relevant payment gateway → system updates the ledger and returns a status.

4.4.3 Functional Requirements

REQ-PAY-1: The system shall allow an authenticated user to initiate a payment to a supported payee.

REQ-PAY-2: The system shall validate a payment request, including sufficient wallet balance, before processing it.

REQ-PAY-3: The system shall require explicit user confirmation before committing a payment.

REQ-PAY-4: The system shall report the final status (success or failure) of a payment to the user.

REQ-PAY-5: The system shall record every payment attempt, whether successful or failed, in the transaction ledger and audit log.

4.5 Transaction History

4.5.1 Description and Priority

Allows an authenticated user to view and filter their past transactions. Priority: Medium.

4.5.2 Stimulus/Response Sequences

- User opens transaction history → system retrieves the user's records → user optionally filters or selects a record → system displays the transaction details.

4.5.3 Functional Requirements

REQ-HIST-1: The system shall store the details and status of every transaction (funding, transfer, and payment).

REQ-HIST-2: The system shall display a user's transaction history on request.

REQ-HIST-3: The system shall allow a user to filter transaction history (TBD: specific filter criteria such as date range, type, or status, see Appendix C).

REQ-HIST-4: The system shall allow a user to view the full details of an individual transaction record.

4.6 Notifications

4.6.1 Description and Priority

Notifies users about transaction and account events as they occur. Priority: Medium.

4.6.2 Stimulus/Response Sequences

- A qualifying event occurs (e.g., a transfer is received, a login occurs from a new device) → the system generates a notification → the notification is delivered to the user through the configured channel.

4.6.3 Functional Requirements

REQ-NOTIF-1: The system shall notify a user when a transaction affecting their wallet completes, whether successfully or unsuccessfully.

REQ-NOTIF-2: The system shall notify a user of significant account events (e.g., verification, password change, login from a new device).

REQ-NOTIF-3: The system shall support delivering notifications through at least one configured channel, such as in-app, email, SMS, or push (TBD: specific channels, see Appendix C).

4.7 Administration

4.7.1 Description and Priority

Allows an administrator to monitor users and transactions and to handle suspicious activity. Priority: Medium.

4.7.2 Stimulus/Response Sequences

- Administrator opens the monitoring dashboard → system displays users/transactions matching the selected criteria → administrator flags or acts on suspicious activity.

4.7.3 Functional Requirements

REQ-ADMIN-1: The system shall allow an administrator to view user accounts and their current status.

REQ-ADMIN-2: The system shall allow an administrator to view and search transactions across all users.

REQ-ADMIN-3: The system shall allow an administrator to flag or restrict an account suspected of suspicious activity.

REQ-ADMIN-4: The system shall log all administrative actions in the audit log.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system shall provide acceptable response times for common operations (e.g., balance lookup, transfer initiation) under expected concurrent load, and shall support a growing number of concurrent users and transaction volume without requiring a redesign of its core architecture.

TBD — quantitative response-time and throughput targets (e.g., "95% of balance queries respond within 2 seconds at N concurrent users") to be finalized with project stakeholders (Section 5.1).

5.2 Safety Requirements

Because the DWMS handles financial funds, the system shall prevent partial or inconsistent updates to wallet balances that could result in loss of funds (see Section 2.5). The system shall maintain a complete audit trail of financial operations to support recovery and investigation in the event of an error or dispute.

TBD — any additional regulatory safety/soundness requirements applicable to digital-wallet operators in the target jurisdiction (Section 5.2).

5.3 Security Requirements

The system shall protect user credentials and personal data using secure authentication and, where applicable, encryption at rest and in transit (see Section 3.4). The system shall enforce authorization checks so that a user can access or modify only their own wallet and transaction data, and so that administrators can perform only permitted administrative actions. The system shall validate all inputs to guard against injection and other common attack classes.

TBD — applicability and target date for formal security certifications (e.g., PCI DSS) (Section 5.3).

5.4 Software Quality Attributes

- Reliability — the system shall maintain accurate wallet and transaction records and handle failures (e.g., a failed external payment call) without corrupting data.
- Availability — the system shall minimize unplanned downtime and provide backup and recovery procedures for its data stores.
- Usability — the system shall offer simple navigation, clear feedback, and understandable error messages (see Section 3.1).
- Scalability — the system's architecture shall support growth in users, transaction volume, and integrations without fundamental redesign.
- Maintainability — the system shall be built with a modular architecture, consistent coding standards, and documentation sufficient to support future modification (see Section 2.5).
- Portability — the client shall operate across the supported browser/OS environments identified in Section 2.4.

5.5 Business Rules

- Only an authenticated user may perform operations on their own wallet.

- A transfer or payment shall not be processed if it would take the sender's balance below zero.
- Transaction limits may vary based on a user's verification status (TBD: specific limit values, see Appendix C).
- Only administrators may access the monitoring/administration functions described in Section 4.7.

6. Other Requirements

Database requirements: the system's data layer shall persist user, wallet, transaction-ledger, and audit-log records with sufficient durability and backup to support recovery (see Sections 2.1, 3.3, and 5.2).

Legal and regulatory requirements: sensitive financial data shall be handled in accordance with applicable data-protection and financial-services regulations for the jurisdictions in which the system operates.

TBD — target jurisdiction(s) and applicable regulatory regime (Section 6).

TBD — internationalization/localization requirements, which are not yet scoped for this release (Section 6).

Appendix A: Glossary

Term	Definition
DWMS	Digital Wallet Management System, the product specified by this document.
Wallet	A user's account balance and associated funding/transfer/payment capabilities within the DWMS.
Ledger	The persistent record of all transactions affecting a wallet.
Audit Log	A tamper-evident record of security- and transaction-relevant events, used for traceability and investigation.
OTP	One-Time Password, used for identity verification.
Payment Gateway	An external service that authorizes and processes payments on the system's behalf.
SRS	Software Requirements Specification.
TBD	To Be Determined — an item requiring a future stakeholder decision, tracked in Appendix C.

Appendix B: Analysis Models

This appendix contains supporting analysis diagrams referenced throughout Sections 2 and 4.

Figure 1 — System Context Diagram

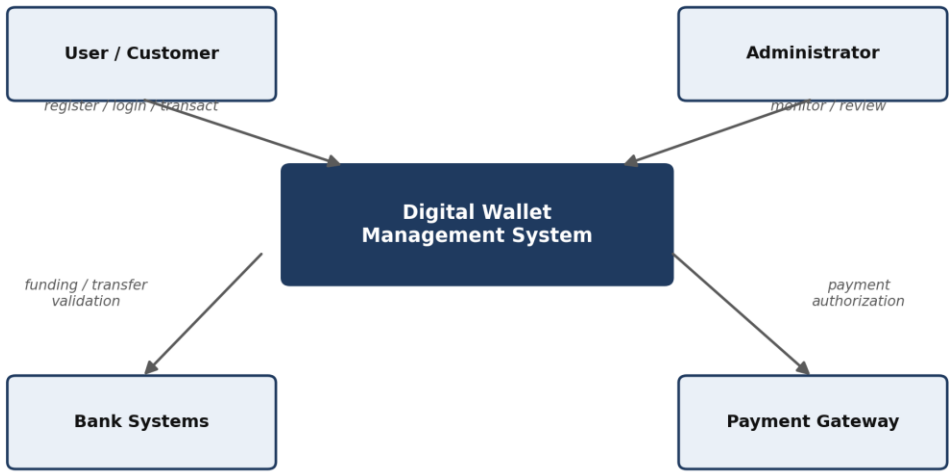


Figure 1 — System Context Diagram: the DWMS and its four external actors (User/Customer, Administrator, Bank Systems, Payment Gateway).

Figure 2 — Layered System Architecture

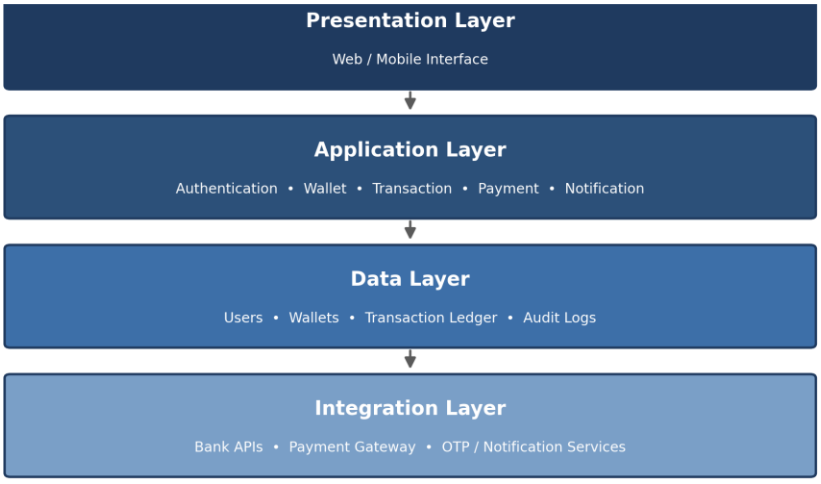


Figure 2 — Layered System Architecture: presentation, application, data, and integration layers, referenced in Sections 2.1 and 2.5.

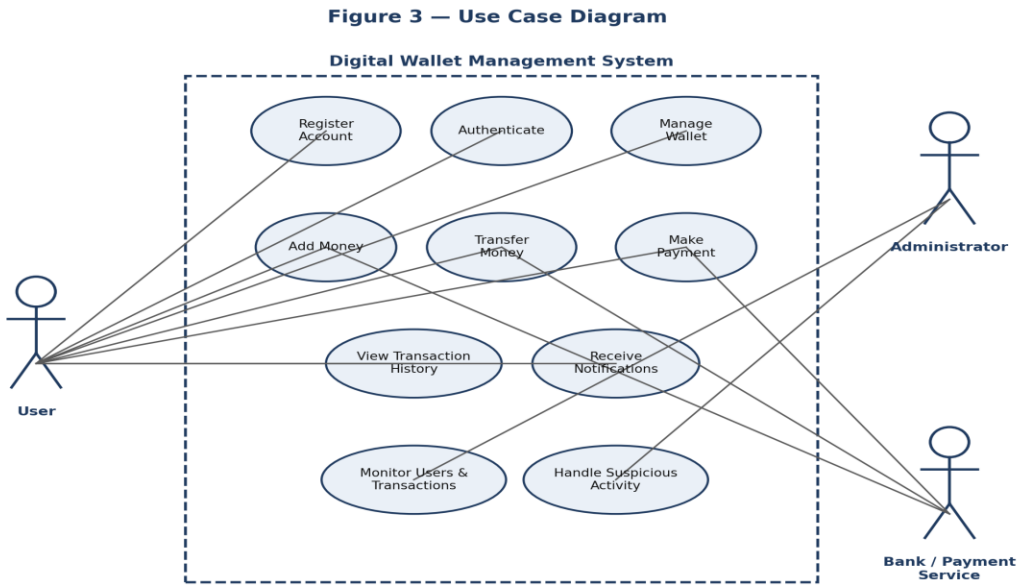


Figure 3 — Use Case Diagram: primary use cases for the User, Administrator, and Bank/Payment Service actors, referenced in Section 2.2.

Appendix C: To Be Determined List

The following items remain open and should be tracked to closure with the appropriate stakeholders before implementation of the affected requirement begins.

TBD-1: Specific supported OS/browser versions and the target cloud hosting environment (Section 2.4).

TBD-2: Applicable regulatory regime for financial/payment data handling (Sections 2.5, 5.3, 6).

TBD-3: User-documentation delivery format (Section 2.6).

TBD-4: Specific vendor products/versions for the database, payment gateway, and notification service (Section 3.3).

TBD-5: Message formats and throughput/rate requirements for external communications (Section 3.4).

TBD-6: Failed-authentication lockout threshold and session-timeout value (Section 4.1).

TBD-7: Transaction-history filter criteria (Section 4.5).

TBD-8: Notification channels to be supported (Section 4.6).

TBD-9: Quantitative performance response-time and throughput targets (Section 5.1).

TBD-10: Specific transaction-limit values by verification tier (Section 5.5).

TBD-11: Target jurisdiction(s) and internationalization scope (Section 6).